



Suppose we are given a graph and are asked to find in it as many independent edges as possible. How should we go about this? Will we be able to pair up all its vertices in this way? If not, how can we be sure that this is indeed impossible? Somewhat surprisingly, this basic problem does not only lie at the heart of numerous applications, it also gives rise to some rather interesting graph theory.

A set M of independent edges in a graph $G = (V, E)$ is called a *matching*. M is a matching of $U \subseteq V$ if every vertex in U is incident with an edge in M . The vertices in U are then called *matched* (by M); vertices not incident with any edge of M are *unmatched*.

matching
matched

A k -regular spanning subgraph is called a k -*factor*. Thus, a subgraph $H \subseteq G$ is a 1-factor of G if and only if $E(H)$ is a matching of V . The problem of how to characterize the graphs that have a 1-factor, i.e. a matching of their entire vertex set, will be our main theme in the first two sections of this chapter.

factor

A generalization of the matching problem is to find in a given graph G as many disjoint subgraphs as possible that are each isomorphic to an element of a given class $\mathcal{H}$ of graphs. This is known as the *packing* problem. It is related to the *covering* problem, which asks how few vertices of G suffice to meet all its subgraphs isomorphic to a graph in $\mathcal{H}$. Clearly, we need at least as many vertices for such a cover as the maximum number k of graphs from $\mathcal{H}$ that we can pack disjointly into G . If there is no cover by just k vertices, perhaps there is always a cover by at most $f(k)$ vertices, where $f(k)$ may depend on $\mathcal{H}$ but not

packing
covering

on G ? In Section 2.3 we shall prove that when $\mathcal{H}$ is the class of cycles, then there is such a function f .

In Section 2.4 we consider packing and covering in terms of edges: we ask how many edge-disjoint spanning trees we can find in a given graph, and how few trees in it will cover all its edges. In Section 2.5 we prove a path cover theorem for directed graphs, which implies the well-known duality theorem of Dilworth for partial orders.

2.1 Matching in bipartite graphs

$G = (V, E)$ For this whole section, we let $G = (V, E)$ be a fixed bipartite graph with
 A, B bipartition $\{A, B\}$. Vertices denoted as a, a' etc. will be assumed to lie
 a, b etc. in A , vertices denoted as b etc. will lie in B .

How can we find a matching in G with as many edges as possible? Let us start by considering an arbitrary matching M in G . A path in G which starts in A at an unmatched vertex and then contains, alternately, edges from $E \setminus M$ and from M , is an *alternating path* with respect to M . Note that the path is allowed to be trivial, i.e. to consist of its starting vertex only. An alternating path P that ends in an unmatched vertex of B is called an *augmenting path* (Fig. 2.1.1), because we can use it to turn M into a larger matching: the symmetric difference of M with $E(P)$ is again a matching (consider the edges at a given vertex), and the set of matched vertices is increased by two, the ends of P .

alternating
path

augment-
ing path

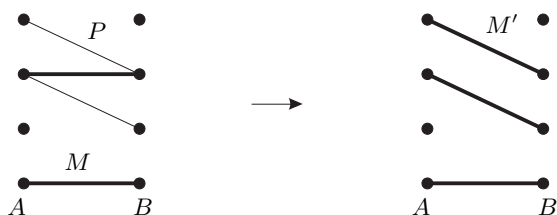


Fig. 2.1.1. Augmenting the matching M by the alternating path P

Alternating paths play an important role in the practical search for large matchings. In fact, if we start with any matching and keep applying augmenting paths until no further such improvement is possible, the matching obtained will always be an optimal one, a matching with the largest possible number of edges (Exercise 1). The algorithmic problem of finding such matchings thus reduces to that of finding augmenting paths – which is an interesting and accessible algorithmic problem.

Our first theorem characterizes the maximal cardinality of a matching in G by a kind of duality condition. Let us call a set $U \subseteq V$ a (*vertex*) *cover* of E if every edge of G is incident with a vertex in U .

cover

Theorem 2.1.1. (König 1931)

The maximum cardinality of a matching in G is equal to the minimum cardinality of a vertex cover of its edges.

Proof. Let M be a matching in G of maximum cardinality. From every edge in M let us choose one of its ends: its end in B if some alternating path ends in that vertex, and its end in A otherwise (Fig. 2.1.2). We shall prove that the set U of these $|M|$ vertices covers E ; since any vertex cover of E must cover M , there can be none with fewer than $|M|$ vertices, and so the theorem will follow. M
 U

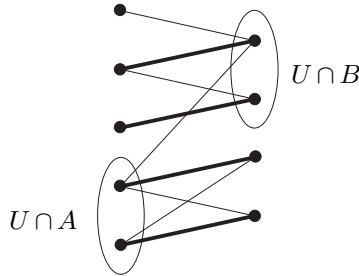


Fig. 2.1.2. The vertex cover U

Note that if an alternating path P ends in a vertex $b \in B$, then $b \in U$: as M is a largest matching, P is not an augmenting path, so b is matched to some $a \in A$ and was put in U when we considered the edge $ab \in M$ while constructing U .

To show that U covers E , let an edge $ab \in E$ be given. If $a \in U$ we are done, so assume that $a \notin U$. To prove $b \in U$, it suffices to show that some alternating path ends in b . If a is unmatched, then ab is such a path. If not, we have $ab' \in M$ for some $b' \in B$. Since $a \notin U$, there exists an alternating path P ending in b' . Depending on whether or not $b \in P$, either Pb or $Pb'ab$ is an alternating path ending in b . □

Let us return to our main problem, the search for some necessary and sufficient conditions for the existence of a 1-factor. In our present case of a bipartite graph, we may as well ask more generally when G contains a matching of A ; this will define a 1-factor of G if $|A| = |B|$, a condition that has to hold anyhow if G is to have a 1-factor.

A condition clearly necessary for the existence of a matching of A is that every subset of A has enough neighbours in B , i.e. that

$$|N(S)| \geq |S| \quad \text{for all } S \subseteq A.$$

marriage
condition

The following *marriage theorem* says that this obvious necessary condition is in fact sufficient:

[2.2.3]

Theorem 2.1.2. (Hall 1935)*G contains a matching of A if and only if $|N(S)| \geq |S|$ for all $S \subseteq A$.*

We give three proofs, of rather different character.¹ In each proof we assume that G satisfies the marriage condition and find a matching of A .

M
 a

First proof. We show that for every matching M of G that leaves a vertex $a \in A$ unmatched there is an augmenting path with respect to M .

Let A' and B' be the sets of vertices in A and B that can be reached by an alternating path from a . Any such path ending at an unmatched $b' \in B'$ is augmenting, so we may assume that all $b' \in B'$ are matched. Their M -neighbours clearly lie in A' , but $a \in A'$ is not among these. Hence by the marriage condition, A' also sends an edge $a'b$ to a vertex $b \notin B'$. Appending this edge to an alternating a - a' path yields an alternating a - b path. This places b in B' , contradicting its choice. $\square$

Second proof. We apply induction on $|A|$. For $|A| = 1$ the assertion is true. Now let $|A| \geq 2$, and assume that the marriage condition is sufficient for the existence of a matching of A when $|A|$ is smaller.

If $|N(S)| \geq |S| + 1$ for every non-empty set $S \subsetneq A$, we pick an edge $ab \in G$ and consider the graph $G' := G - \{a, b\}$ obtained by deleting its ends. Then every non-empty set $S \subseteq A \setminus \{a\}$ satisfies

$$|N_{G'}(S)| \geq |N_G(S)| - 1 \geq |S|,$$

so by the induction hypothesis G' contains a matching of $A \setminus \{a\}$. Together with the edge ab , this yields a matching of A in G .

Suppose now that A has a non-empty proper subset A' with $|B'| = |A'|$ for $B' := N(A')$. By the induction hypothesis, $G' := G[A' \cup B']$ contains a matching of A' . But $G - G'$ satisfies the marriage condition too: for any set $S \subseteq A \setminus A'$ with $|N_{G-G'}(S)| < |S|$ we would have $|N_G(S \cup A')| < |S \cup A'|$, contrary to our assumption. Again by induction, $G - G'$ contains a matching of $A \setminus A'$. Putting the two matchings together, we obtain a matching of A in G . $\square$

H

For our last proof, let H be an edge-minimal subgraph of G that satisfies the marriage condition and contains A . Note that $d_H(a) \geq 1$ for every $a \in A$, by the marriage condition with $S = \{a\}$.

Third proof. We show that $d_H(a) = 1$ for every $a \in A$. The edges of H then form a matching of A , since by the marriage condition no two such edges can share a vertex in B .

Suppose a has distinct neighbours b_1, b_2 in H . By definition of H , the graphs $H - ab_1$ and $H - ab_2$ violate the marriage condition. So for

¹ The theorem can also be derived easily from König's theorem; see Exercise 5.

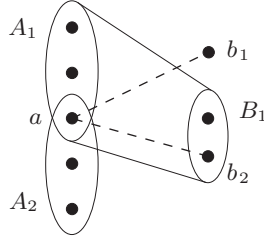


Fig. 2.1.3. B_1 contains b_2 but not b_1

$i = 1, 2$ there is a set $A_i \subseteq A$ containing a such that $|A_i| > |B_i|$ for $B_i := N_{H-ab_i}(A_i)$ (Fig. 2.1.3). Since $b_1 \in B_2$ and $b_2 \in B_1$,

$$\begin{aligned}
 |N_H(A_1 \cap A_2 \setminus \{a\})| &\leq |B_1 \cap B_2| \\
 &= |B_1| + |B_2| - |B_1 \cup B_2| \\
 &= |B_1| + |B_2| - |N_H(A_1 \cup A_2)| \\
 &\leq |A_1| - 1 + |A_2| - 1 - |A_1 \cup A_2| \\
 &= |A_1 \cap A_2| - 2 \\
 &= |A_1 \cap A_2 \setminus \{a\}| - 1.
 \end{aligned}$$

Hence H violates the marriage condition, contrary to assumption. $\square$

This last proof has a pretty ‘dual’, which begins by showing that $d_H(b) \leq 1$ for every $b \in B$. See Exercise 6 and its hint for details.

Corollary 2.1.3. *Every k -regular ($k \geq 1$) bipartite graph has a 1-factor.*

Proof. If G is k -regular, then clearly $|A| = |B|$; it thus suffices to show by Theorem 2.1.2 that G contains a matching of A . Now every set $S \subseteq A$ is joined to $N(S)$ by a total of $k|S|$ edges, and these are among the $k|N(S)|$ edges of G incident with $N(S)$. Therefore $k|S| \leq k|N(S)|$, so G does indeed satisfy the marriage condition. $\square$

In some real-life applications, matchings are not chosen on the basis of global criteria for the entire graph but evolve as the result of independent decisions made locally by the participating vertices. A typical situation is that vertices are not indifferent to which of their incident edges are picked to match them, but prefer some to others. Then if M is a matching and $e = ab$ is an edge not in M such that both a and b prefer e to their current matching edge (if they are matched), then a and b may agree to change M locally by including e and discarding their earlier matching edges. The matching M , although perhaps of maximum size, would thus be *unstable*.

preferences
stable
matching

More formally, call a family $(\leq_v)_{v \in V}$ of linear orderings $\leq_v$ on $E(v)$ a *set of preferences* for G . Then call a matching M in G *stable* if for every edge $e \in E \setminus M$ there exists an edge $f \in M$ such that e and f have a common vertex v with $e <_v f$. The following result is sometimes called the *stable marriage theorem*; see Exercises 16 and 17 for a discussion of alternative proofs.

[5.4.4]

Theorem 2.1.4. (Gale & Shapley 1962)

For every set of preferences, G has a stable matching.

Proof. Call a matching M in G *better* than a matching $M' \neq M$ if M makes the vertices in B happier than M' does, that is, if every vertex b in an edge $f' \in M'$ is incident also with some $f \in M$ such that $f' \leq_b f$. We shall construct a sequence of better and better matchings. Since these can increase the happiness of a fixed vertex b at most $d(b)$ times, this process will terminate.

Given a matching M , call a vertex $a \in A$ *acceptable* to $b \in B$ if $e = ab \in E \setminus M$ and any edge $f \in M$ at b satisfies $f <_b e$. Call $a \in A$ *happy* with M if a is unmatched or its matching edge $f \in M$ satisfies $f >_a e$ for all edges $e = ab$ such that a is acceptable to b .

Starting with the empty matching, let us construct a sequence of matchings that keep all the vertices in A happy. Given such a matching M , consider a vertex $a \in A$ that is unmatched but acceptable to some $b \in B$. (If no such a exists, terminate the sequence.) Add to M the $\leq_a$ -maximal edge ab such that a is acceptable to b , and discard from M any other edge at b .

Clearly, each matching in our sequence is better than the previous and keeps the vertices in A happy (which they initially are, when $M = \emptyset$). So the sequence continues until it terminates with a matching M such that no unmatched vertex in A is acceptable to any of its neighbours in B . As every matched vertex in A is happy with M , this matching is stable. $\square$

Despite its seemingly narrow formulation, the marriage theorem counts among the most frequently applied graph theorems, both outside graph theory and within. Often, however, recasting a problem in the setting of bipartite matching requires some clever adaptation. As a simple example, we now use the marriage theorem to derive one of the earliest results of graph theory, a result whose original proof is not all that simple, and certainly not short:

Corollary 2.1.5. (Petersen 1891)

Every regular graph of positive even degree has a 2-factor.

Proof. Let G be any $2k$ -regular graph ($k \geq 1$), without loss of generality connected. By Theorem 1.8.1, G contains an Euler tour $v_0 e_0 \dots e_{\ell-1} v_\ell$, with $v_\ell = v_0$. We replace every vertex v by a pair (v^-, v^+) , and every edge $e_i = v_i v_{i+1}$ by the edge $v_i^+ v_{i+1}^-$ (Fig. 2.1.4). The resulting bipartite graph G' is k -regular, so by Corollary 2.1.3 it has a 1-factor. Collapsing every vertex pair (v^-, v^+) back into a single vertex v , we turn this 1-factor of G' into a 2-factor of G . $\square$ (1.8.1)

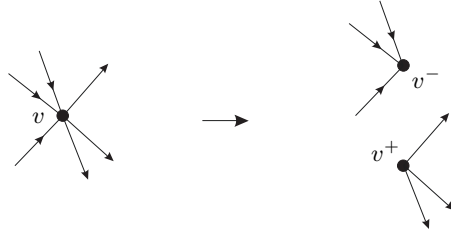


Fig. 2.1.4. Splitting vertices in the proof of Corollary 2.1.5

2.2 Matching in general graphs

Given a graph G , let us denote by $\mathcal{C}_G$ the set of its components, and by $q(G)$ the number of its *odd components*, those of odd order. If G has a 1-factor, then clearly

$$q(G - S) \leq |S| \quad \text{for all } S \subseteq V(G),$$

$\mathcal{C}_G$
 $q(G)$

Tutte's condition

since every odd component of $G - S$ will send a factor edge to S .

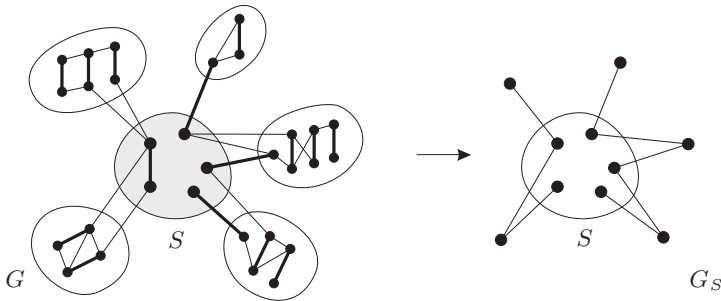


Fig. 2.2.1. Tutte's condition $q(G - S) \leq |S|$ for $q = 3$, and the contracted graph G_S from Theorem 2.2.3.

Again, this obvious necessary condition for the existence of a 1-factor is also sufficient: